Calc II Section Notes

Fall 2026

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Contents

Chapter 1

Applications of Integrals

Week 1 (8/31-9/3)

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Problem 1.1 (discuss with your teammates). You are an open water diver, which of the following sharks would you be excited to see? Rank from 1 to 5 most excited to least excited.

Shark Type	**Diet**	**Humans**	**Appearance**
Tiger shark	Sea turtles, rays, other sharks, fish, seabirds, dolphins, squid, crustaceans, and carrion	142 (39 fatal)	Long body with a broad head and dark vertical bands along the sides.
Hammerhead shark	Fish, squid, crustaceans, rays, and smaller sharks	18 (0 fatal)	Distinctive hammer-shaped head with eyes positioned far apart.
Nurse shark	Fish, stingrays, octopus, squid, clams, and crustaceans	9 (0 fatal)	Rounded head with small eyes and two barbels near the mouth.
Thresher shark	Mostly schooling fish such as herring and mackerel; also squid	0 listed	Extremely long upper tail fin, nearly as long as the rest of the body.
Great white shark	Fish, rays, other sharks, squid, and marine mammals	351 (59 fatal)	Muscular body with prominent triangular teeth. Powerful and heavy-looking.

Problem 1.2. Approximate the area under $f(x) = x^2$ on $[0, 1]$ using Riemann sums with $N = 2, 3$ and some choice of sample points. Then think about the limit as $N \to \infty$.

Proof. We will use left endpoints. When $N = 2$, $x_1 = 0, x_2 = 1/2$. Thus

$$R_2 = 0 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

When $N = 3$, $x_1 = 0, x_2 = 1/3, x_3 = 2/3$. Then

$$R_3 = 0 + \frac{1}{9} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{3} = \frac{5}{27}$$

Now we compute when $N \to \infty$

$$R = \lim_{N\to\infty} \sum_{i=0}^{N-1} \left(\frac{i}{N}\right)^2 \frac{1}{N} = \frac{1}{6}$$

$\square$

Problem 1.3. Compute and interpret the limit

$$\lim_{N\to\infty} \sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right)$$

Hint:

$$\sum_{i=1}^{N} i = \frac{N(N+1)}{2}, \quad \sum_{i=1}^{N} i^2 = \frac{N(N+1)(2N+1)}{6}$$

Proof. Using the formulas in the hint, one can directly compute the limit to be $1/6$. Now we can also write it as a Riemann sum:

$$\sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right) = \sum_{i=1}^{N} \frac{1}{N} \left(\frac{i}{N} - \frac{i^2}{N^2}\right)$$

Then $\Delta x = \frac{1}{N}$, and $x_i = \frac{i}{N}$. Letting $N \to \infty$, this is the Riemann sum of the following

$$\int_0^1 (x - x^2)\, dx$$

$\square$

Problem 1.4. Compute

$$\int_{-1}^{5} (1 + 3x)\, dx$$

using Riemann sums.

Proof. We divide the interval $[-1, 5]$ into N intervals, then $\Delta x = 6/N$, and if we use right endpoints, the ith sample point is $x_i = -1 + 6i/N$. Then

$$\sum_{i=1}^{N} f(x_i)\Delta x = \sum_{i=1}^{N} (-2 + 18i/N) \frac{6}{N}$$

letting $N \to \infty$, the answer is 42. $\square$